

Utilization of Electricity 2: Energy Efficiency Practices

Introduction: Less Energy, Same Benefits — The Promise of Efficiency

Every time a light is left on in an empty room, a charger remains plugged in without a device attached, or an old refrigerator hums inefficiently in a hot kitchen, energy is being wasted. These small acts of carelessness, repeated across millions of households daily, add up to staggering losses — wasted money, wasted resources, and unnecessary harm to the environment.

Energy efficiency means using less energy to perform the same task. An energy-efficient LED bulb produces the same amount of light as an incandescent bulb but uses approximately 75% less electricity. An energy-efficient refrigerator keeps food cold while consuming significantly less power than an older model. Efficiency is not about doing without — it is about doing more with less.

Energy conservation goes beyond efficiency. It involves changing behavior to reduce energy use altogether — turning off lights when leaving a room, unplugging devices not in use, and using natural light instead of artificial lighting. Together, efficiency and conservation form the foundation of responsible energy use.

This reading material examines the principles of energy efficiency, common sources of energy waste, practical strategies for reducing consumption at home and in school, and the broader implications of energy conservation for the environment, the economy, and future generations.

Part 1: Understanding Energy Efficiency and Conservation

Definitions

Term	Definition
Energy efficiency	Using less energy to perform the same task or produce the same output
Energy conservation	Changing behavior to reduce energy use, such as turning off lights when leaving a room
Energy waste	Energy that is consumed but does not produce useful work or benefit

Key Distinction

Concept	Focus	Example
Energy efficiency	Technology and equipment	Replacing an incandescent bulb with an LED bulb
Energy conservation	Behavior and habits	Turning off lights when leaving a room
Energy waste	Inefficient use or unnecessary consumption	Leaving a computer on overnight

Why Energy Efficiency Matters

Reason	Explanation
Reduces electricity bills	Lower consumption means lower monthly costs
Protects the environment	Less electricity generation means fewer greenhouse gas emissions
Conserves natural resources	Reduces demand for coal, natural gas, and other finite resources
Improves energy security	Reduces dependence on imported fuels
Extends equipment life	Efficient operation reduces wear and tear

Part 2: Common Sources of Energy Waste

At Home

Source of Waste	Description	Estimated Waste
Lights left on in empty rooms	Occupants forget to turn off lights when leaving	Significant
Devices left on standby	Televisions, computers, and chargers consume power even when not in use ("phantom load")	5-10% of household electricity
Inefficient lighting	Incandescent bulbs waste 90% of energy as heat	Very high

Refrigerator door left open	Cold air escapes; compressor works harder to maintain temperature	Moderate
Overcooling or overheating	Setting thermostats too high or too low wastes energy	High
Old, inefficient appliances	Older models consume significantly more energy than ENERGY STAR rated models	Very high

At School

Source of Waste	Description
Lights left on in empty classrooms	Rooms are lit even when unoccupied
Windows covered when natural light is available	Artificial lighting used unnecessarily
Electric devices left on when not in use	Projectors, computers, and fans running in empty rooms
Overlit areas	More lighting than necessary for the task
Inefficient lighting	Use of incandescent or fluorescent tubes instead of LEDs
Electric fans running unnecessarily	Fans left on in empty or naturally ventilated rooms
Doors/windows left open while AC is on	Cooled air escapes; AC works harder
Improper maintenance of ventilation	Unclean fans or AC filters reduce efficiency

In Businesses and Industries

Source of Waste	Description
Equipment left running during breaks	Machines continue to consume power when not in use
Compressed air leaks	Significant energy loss in manufacturing facilities
Poor insulation	Heat loss in winter; cool air loss in summer
Inefficient lighting	Use of outdated lighting technology
Uncontrolled HVAC systems	Heating and cooling running when spaces are unoccupied

Part 3: Practical Strategies for Energy Efficiency

Lighting

Strategy	How It Saves Energy
Replace incandescent bulbs with LEDs	LEDs use 75% less energy and last 25 times longer
Turn off lights when leaving a room	Eliminates unnecessary consumption
Use natural light when possible	Reduces need for artificial lighting
Install motion sensors	Lights automatically turn off in unoccupied rooms
Use task lighting instead of overhead lighting	Illuminates only the area being used

Comparison of Light Bulbs:

Type	Energy Use	Lifespan	Efficiency
Incandescent	High (60-100W)	1,000 hours	Low (90% lost as heat)
CFL (compact fluorescent)	Moderate (13-15W)	8,000 hours	Moderate
LED (light-emitting diode)	Low (8-12W)	25,000 hours	High

Appliances and Electronics

Strategy

How It Saves Energy

Unplug devices when not in use

Eliminates phantom load (power consumed even when off)

Use power strips

One switch turns off multiple devices at once

Choose ENERGY STAR rated appliances

Certified to meet strict energy efficiency guidelines

Run full loads in dishwasher and washing machine

Maximizes energy per cycle

Air-dry dishes and clothes when possible

Eliminates energy used by drying cycles

Clean refrigerator coils

Dirty coils reduce efficiency; compressor works harder

Heating, Ventilation, and Air Conditioning (HVAC)

Strategy	How It Saves Energy
Set thermostat appropriately	20-22°C (68-72°F) in winter; 24-26°C (75-78°F) in summer
Use ceiling fans instead of AC	Fans use much less energy than air conditioners
Close doors and windows when AC is on	Prevents cooled air from escaping
Clean AC filters regularly	Dirty filters reduce efficiency and increase energy use
Use natural ventilation when weather permits	Open windows instead of running AC or fans
Seal gaps around doors and windows	Prevents drafts and reduces heating/cooling load

Behavioral Changes

Habit	Energy Saved
Turning off lights when leaving a room	Up to 10% of lighting energy
Unplugging phone charger after use	Eliminates phantom load (0.5-2W per charger)
Using natural light during daytime	100% of lighting energy during daylight hours
Taking shorter showers	Reduces water heating energy
Washing clothes in cold water	Eliminates energy used to heat water
Hanging laundry to dry instead of using a dryer	100% of dryer energy saved

Part 4: The Philippine Energy Context

Energy Consumption Trends

According to Philippine power statistics, energy consumption has steadily increased over the past two decades. The major consuming sectors are:

Sector	Share of Consumption	Primary Uses
Residential	Significant	Lighting, appliances, air conditioning, entertainment
Commercial	Significant	Lighting, HVAC, office equipment, elevators
Industrial	Largest	Manufacturing, machinery, processing

Energy Sources in the Philippines

Source	Share	Characteristics
Coal	Largest	High carbon emissions; non-renewable
Natural gas	Significant	Lower emissions than coal; non-renewable
Renewable (geothermal, hydro, solar, wind)	Growing	Clean; sustainable; no fuel cost
Oil	Small	Expensive; used primarily for transportation

Why Energy Efficiency Matters in the Philippines

Reason	Explanation
High electricity costs	The Philippines has among the highest electricity rates in Asia
Limited domestic energy resources	Relies on imported fuels, affecting energy security
Vulnerability to climate change	Reducing emissions helps mitigate climate impacts
Growing energy demand	Economic growth increases electricity consumption
Environmental protection	Reduced generation means less air and water pollution

Part 5: Energy Efficiency Standards and Labeling

What Is Energy Labeling?

Energy labels provide consumers with information about an appliance's energy consumption and efficiency. In the Philippines, the Department of Energy (DOE) implements an energy labeling program for appliances such as refrigerators, air conditioners, and televisions.

What to Look For on an Energy Label

Information	What It Means
Energy efficiency rating	How efficient the appliance is compared to others in its class
Annual energy consumption	Estimated kilowatt-hours (kWh) used per year

Information	What It Means
Energy cost estimate	Estimated annual operating cost
Comparative scale	Shows where the appliance ranks (e.g., 1 to 5 stars)

Choosing Energy-Efficient Appliances

Step	Action
1. Check the energy label	Compare efficiency ratings among models
2. Look for ENERGY STAR certification	International standard for energy efficiency
3. Consider lifetime cost, not just purchase price	An efficient appliance may cost more initially but saves money over time
4. Size the appliance appropriately	An oversized appliance wastes energy
5. Maintain appliances properly	Regular maintenance preserves efficiency

Part 6: Republic Act 11285 — Energy Efficiency and Conservation Act

Overview of RA 11285

Republic Act No. 11285, also known as the **Energy Efficiency and Conservation Act**, was enacted in 2019 to institutionalize energy efficiency and conservation as a national priority. The law aims to promote the efficient use of energy, reduce energy consumption, and encourage investments in energy-saving technologies.

Key Provisions

Provision	Description
Declaration of Policy	Establishes energy efficiency and conservation as a national priority
Energy Efficiency Standards	Mandates minimum energy performance standards for appliances and equipment
Energy Labeling	Requires energy labels on appliances to inform consumers
Government Leadership	Requires government agencies to implement energy efficiency measures
Incentives	Provides tax incentives for energy efficiency projects
Energy Audits	Requires large commercial and industrial entities to conduct regular energy audits

How RA 11285 Affects Households

Requirement	Implication
Energy labels on appliances	Consumers can compare efficiency before purchasing
Minimum efficiency standards	Inefficient appliances cannot be sold
Government information campaigns	Public becomes more aware of energy-saving practices

How RA 11285 Affects Businesses and Industries

Requirement	Implication
Mandatory energy audits	Large energy consumers must identify efficiency opportunities
Energy efficiency projects	Businesses are encouraged to invest in energy-saving technologies
Reporting requirements	Designated establishments must report energy consumption and conservation measures

Part 7: Energy Efficiency and the Sustainable Development Goals (SDGs)

Energy efficiency directly contributes to several United Nations Sustainable Development Goals.

SDG	Connection to Energy Efficiency
SDG 7: Affordable and Clean Energy	Efficiency makes energy more affordable and reduces the need for additional generation
SDG 12: Responsible Consumption and Production	Efficiency reduces resource consumption and waste
SDG 13: Climate Action	Efficiency reduces greenhouse gas emissions
SDG 11: Sustainable Cities and Communities	Efficient buildings and transportation reduce urban energy demand

Question to Ponder

How does saving energy at home and school contribute to these global goals?

Action	Contributes to SDG
Turning off lights when leaving a room	SDG 12 (responsible consumption)
Using energy-efficient appliances	SDG 7 (affordable and clean energy)
Reducing electricity consumption	SDG 13 (climate action)
Promoting energy conservation awareness	SDG 11 (sustainable communities)

Part 8: Conducting an Energy Audit

What Is an Energy Audit?

An **energy audit** is a systematic inspection and analysis of energy use in a building or facility. It identifies where energy is being wasted and recommends measures to improve efficiency.

Simple Home Energy Audit Checklist

Area	What to Check
Lighting	Are incandescent bulbs still in use? Are lights left on in empty rooms?
Appliances	Are old, inefficient appliances still in use? Are devices left on standby?
Refrigerator	Is the door seal tight? Are coils clean? Is the temperature set appropriately?

Area	What to Check
Air conditioning	Are filters clean? Are doors and windows closed when AC is on? Is the temperature set at 24-26°C?
Electronics	Are computers, televisions, and chargers unplugged when not in use?
Water heating	Is the water heater thermostat set appropriately? Are showers kept short?

Energy Audit Data Table

Appliance/Device	Number of Devices	Wattage	Hours Used per Day	Energy per Day (kWh)	Energy-Saving Action
Electric fan	4	75 W	8	2.4 kWh	Use natural ventilation when possible
Refrigerator	1	150 W	24	3.6 kWh	Keep coils clean; set temperature properly
Television	2	100 W	4	0.8 kWh	Turn off when no one is watching
LED bulbs	10	10 W	5	0.5 kWh	Turn off when leaving the room

Key Terms

Term	Definition
Energy efficiency	Using less energy to perform the same task
Energy conservation	Changing behavior to reduce energy use
Energy waste	Energy consumed that does not produce useful work
Phantom load (standby power)	Electricity consumed by devices even when turned off but still plugged in
ENERGY STAR	International standard for energy-efficient products
Energy audit	Systematic inspection of energy use to identify waste and improvement opportunities
RA 11285	The Energy Efficiency and Conservation Act of the Philippines
LED (Light Emitting Diode)	Highly efficient lighting technology
Sustainable Development Goals (SDGs)	United Nations global goals for sustainable development

Review Questions

Responses should be written on a separate sheet of paper.

1. Distinguish between **energy efficiency** and **energy conservation**. Provide one example of each.
2. A homeowner receives an unusually high electricity bill. List three possible sources of energy waste and describe one solution for each.
3. Why are LED bulbs preferred over incandescent bulbs for energy efficiency? Explain using specific data.
4. What is **phantom load**? Which household devices are most likely to contribute to phantom load, and how can it be reduced?
5. Explain the purpose of **RA 11285 (Energy Efficiency and Conservation Act)**. How does this law affect consumers purchasing new appliances?